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The ghost of the past anthropogenic impact: Reef-decapods as bioindicators of threatened marine ecosystems

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A B S T R A C T

This study demonstrated the plasticity of reef-decapod according with the exposed anthropogenic pressure. Comparing the reef-decapods at sites with different anthropogenic pressures in two different ecoregions (tropical and subtropical reef ecotypes), we demonstrated that historical anthropogenic activity is molding directly and indirectly their population. A direct impact over target decapods for fishing (e.g. Lobsters and large crabs) that are removed directly by human activities. A positive indirect impact related to a top-down trophic imbalance with the increase of naturally dominant preys (small decapod) due to the removal of top predators. Thus, insufficient predators to control the abundant prey in a classic unbalanced prey-predator relationship. An indirect and negative impact, related to a bottom-up trophic imbalance, is the exclusion of niche-restricted species in association with benthic cover (e.g. corals). This also has the positive outcome, however, of increasing numbers of herbivores, detritivores and scavengers. As hypothesis, we propose that a current biodiversity balance in anthropogenically impacted ecosystem is just reflecting past trophic cascade events. Where the remaining biodiversity is monopolizing the vacant niches after the removal of components in a trophic chain. Therefore, we are suggesting that reef-decapod biodiversity, observed using visual identification, can be used to highlight the "Ghost of the past anthropogenic impact" in threatened ecosystems.

1. Introduction

In the current Anthropocene, historical human pressure threatens marine ecosystems worldwide, resulting in defaunation, loss of biological functionality, and the systemic collapse of ecosystems and natural resources (Giacomini and Galetti, 2013). In an ecological point of view, the anthropogenic activities can affect trophic levels in the ecosystem, resulting in cascading effects (Boudreau and Worm, 2012; Galetti and Dirzo, 2013; Jackson et al., 2001; Maitra et al., 2018) with top-down effects, such as an increase in prey (or plants) with the removal of predators, or bottom-up effects, such as how herbivorous and carnivorous species can be indirectly removed from an ecosystem after the forced removal of photosynthetic organisms (such as coral and plants) (Jackson et al., 2001; Lynam et al., 2017; Maitra et al., 2018). Indeed, several human activities have been removing components of ecosystems. The historical fishing activities, have been removing top predators in the marine trophic chain worldwide (Boudreau and Worm, 2012; Jackson et al., 2001; Lynam et al., 2017; Maitra et al., 2018; Randall and Bishop, 1967; Segal et al., 2017). Tourism have been modifying the benthic coverage on reefs ecosystems (Barradas et al., 2012; Dwyer and Edwards, 2000). The excessive input of organic matter and nitrogen compounds, increases the coastal eutrophication (Barradas et al., 2012; Moss, 2011), leading to an increase in algae (including microalgae and phytoplankton) and a consequent increase in the primer consumers (detritivores and herbivores), modifying the benthic community (Grall and Chauvaud, 2002).

The growing population in the coastal urban centers and the tireless Tragedy of Commons (Hardin, 1968), created fragmented ecosystems

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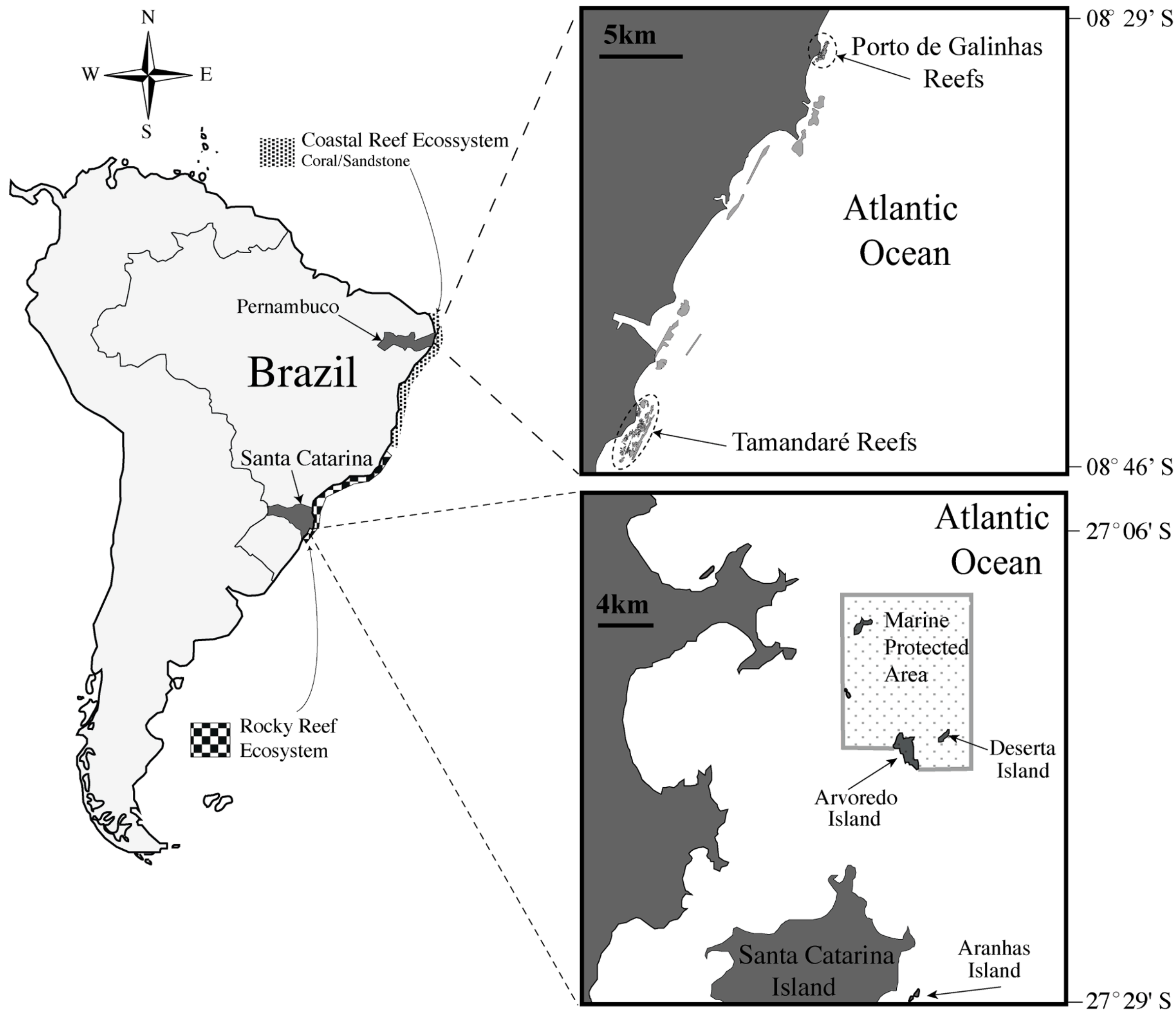

Fig. 1. Map illustrating the locations of the study areas, and the specific locations of the study sites in Santa Catarina (rock shores), and Pernambuco (coastal coral reefs).

worldwide. Indeed, there is a threatened gradient according to the proximity of large urban centers (Mirabella and Allacker, 2017), related to the accessibility of the resource; that is, how far, how protected, or how difficult it is for humans to access the resource determines how threatened it is (Al Maslamani et al., 2018; Jackson et al., 2001; Pereira et al., 2018). In this scenario of fragmented ecosystems there is a virtual absence of historical information regarding the removed components in the trophic chains, suggesting a need for new environmental assessment approaches that highlight past anthropogenic impacts which have not yet been sufficiently described.

A dismissed biodiversity in aquatic environmental assessment and monitoring is the reef-decapods (decapods associated to hard-substrates). They partake important ecological roles in all marine ecosystems (Boudreau and Worm, 2012; Watling et al., 2013), presenting specific assemblages zoned per habitat in reef ecosystems, such as external, crevicular (cryptic) and peripherical soft bottom (Giraldes et al., 2017) and houses one of the greatest diversity of marine species in the sea (De Grave et al., 2009). In addition, Reef-decapods are referred to as the "cleaning crew" in aquariums, responsible for the maintenance of those closed marine ecosystems, partaking same functions in natural environments. Functions of decapod species include natural pest-control (Calado and Narciso, 2005); natural seaweed control (Butler and Mojica, 2012; Coen, 1988); natural bioengineers (bioturbation), (Al-khayat and Giraldes, 2020; Norling et al., 2007); natural cleaners, cleaning and removing parasites and dead tissues of coral and fish (Glynn and Enochs, 2011; Stewart et al., 2006; Watling et al., 2013); and they are symbiotic organisms with close association with several species (Glynn and Enochs, 2011; Stewart et al., 2006; Thompson, 2004). Highlighting that decapods are the main food resource (prey) for several fish species, presenting representative species in most trophic chains in the sea (Boudreau and Worm, 2012; Pereira et al., 2018; Randall and Bishop, 1967); they are also efficient herbivores (Butler and Mojica, 2012), scavengers and detritivores (Boudreau and Worm, 2012; Grall and Chauvaud, 2002; Ramsay et al., 1997; Witte, 1999). Therefore, considering their ecological importance and that the methodology for underwater monitoring have been well stablished (Gaeta et al., 2011; Giraldes et al., 2017, 2015a, 2015b, 2012) it is possible to assume that the underwater monitoring of decapods has great potential to enhance the understanding of ecological imbalances caused by anthropogenic actions.

Drawing an analogy with the ecological concept of the ghost of the competition past (Connell, 1980) it is possible to assume that the human being in large coastal urban centers (in supper population rates), represents a competitor in the natural ecosystems, which have been removing for decades target components in the marine trophic chain (Jackson et al., 2001; Pereira et al., 2018). Considering that the remaining biodiversity monopolizes the vacant niches after the removal of components in a trophic chain, it is possible to assume that in areas with permanent and historical anthropogenic pressure a new stablished ecosystem will take place. In this study, we are hypothesizing that reef-

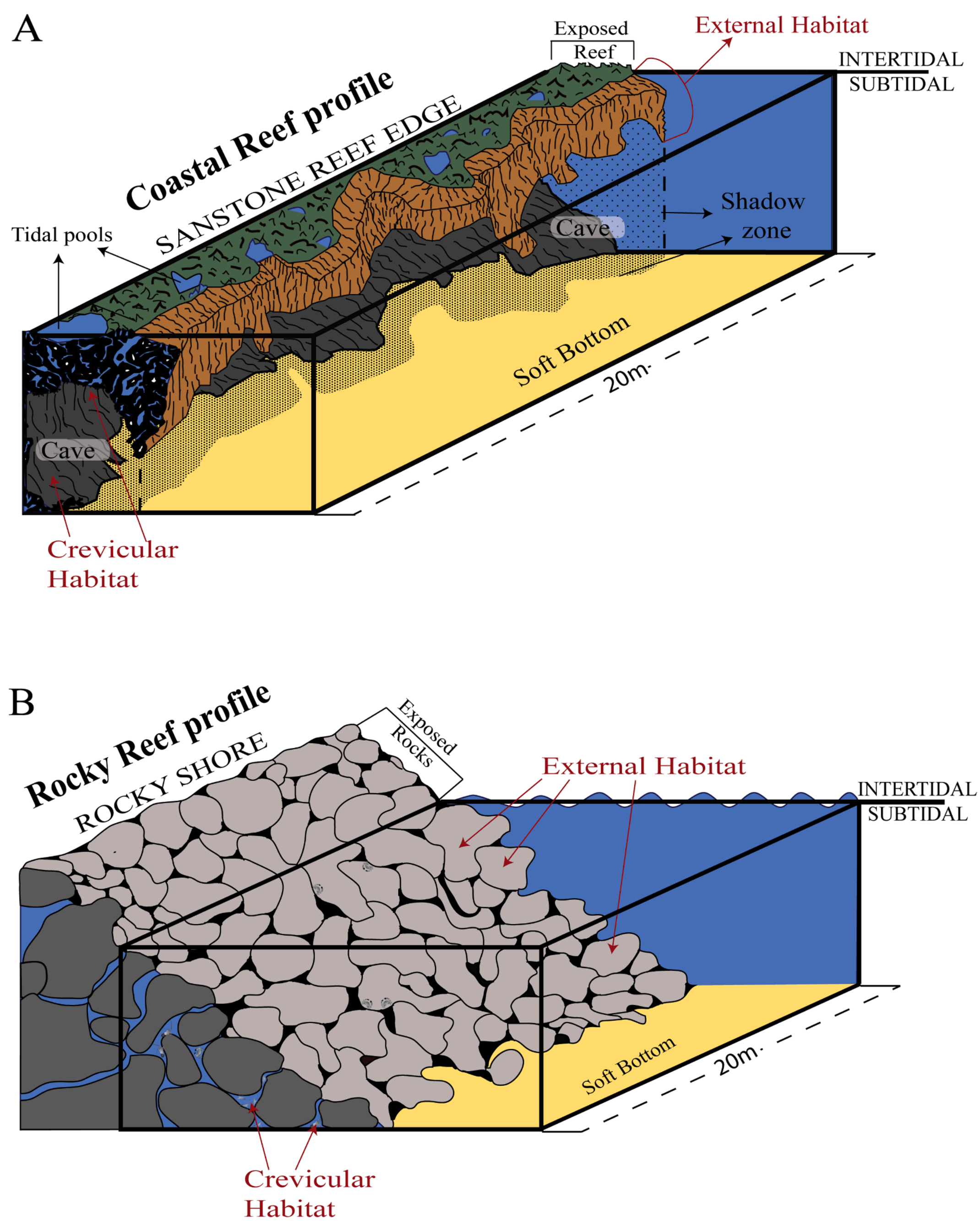

Fig. 2. Illustration of the peculiar features of the hard substrate in the study sites: (A) coastal reefs edge and (B) rocky shores. Crevicular and external habitats are highlighted.

decapods can be used as a target group for highlighting the unmeasured past anthropogenic impacts in a threatened ecosystem. With the population of target reef-decapods reflecting the past imbalance, disappearing from the ecosystems, or occupying a vacant niche due the absence of predators and/or the excessive food resource. Owing to the absence of real pristine ecosystems, in this study we compared the reef-decapod biodiversity at sites with different anthropogenic pressures within the two main representative reef ecotypes in Brazil, the tropical coastal reefs in the northeast shoreline and the subtropical rocky reef in the southeast waters. Here we are presenting an ecological discussion about the natural re-modeling of reef-decapods population when under anthropogenic pressures.

2. Material and methods

2.1. Study areas

The two main reef seascapes in Brazil were studied here (Fig. 1) the tropical coastal coral reef ecosystem with a peculiar coastal sandstone formation (from ~ 5 S to 18 S) and the subtropical rocky reef ecosystem (from ~ 20 S to 28 S) (Aued et al., 2018; Leão et al., 2003). Housed in two different ecoregions named as “Tropical Southwestern Atlantic” and “Warm Temperate Southwestern Atlantic” respectively (Spalding et al., 2007).

Two coastal coral reefs (~8 S) were selected, the reef bench of Porto de Galinhas (Po.Ga.) (8°30′07″ S, 35°00′08″ W) and the reef bench of Tamandare (Tam.) (08° 44′ 23″S, 35° 02′ 28″ W). The Po.Ga. reef bench is the main tourist attraction of a large tourist industry in the region, and is close (50 km) to Recife city, the region's largest urban center (~4 million inhabitants), while Tam. is more than 100 km from Recife and provides only very poor-quality roads for accessing its reef area. In Po. Ga., after a long fishing history, the local fishers migrated to the more profitable tourism industry that took over the exploitation of the reef. After this migration, recreational spearfishing and fishing with rods still occurred and a massive daily artificial fish feeding was implemented (Engmann, 2006; Feitosa et al., 2012). Conversely, the Tam. reef bench houses a MPA coral reef area and it is in a small fishing village in front of the local governmental body for nature protection (CEPENE-ICMBIO) that cares and manage the activities within the Coral Coast area (Ferreira et al., 2001). Tam. reef has less tourism, but allows extensive artisanal fishing, particularly lobster fishing using gillnets in the surrounded MPA area (Giraldes et al., 2015b) the main income for this small fishing village. Regarding the benthic coverage, because of the mass tourism that Po.Ga. experiences, there are significantly more bare rocks and less live coverage (corals, zoanthids and algae) than at the Tamandare reefs (Barradas et al., 2012; de Santana et al., 2015; Engmann, 2006; Santos et al., 2015). In addition, levels of organic matter (Ammonia, Nitrate and Phosphate) are significantly increased in Po.Ga. on account of the massive artificial fish feedings on a daily basis, the trampling of benthic coverage, sewage discharge that goes directly into the seawater, and large quantities of trash associated with uncontrolled tourism (Barradas et al., 2012; Engmann, 2006; Leão et al., 2003; Santos et al., 2015).

The studied rocky reef sites (~27 S) were in coastal Islands, Arvoredo (Arvo.) (27° 17′ 11″ S, 48° 22′ 00″ W), Deserta (Dese.) (27° 16′ 00″ S, 48° 20′ 00″ W) and Aranhas (Aran.) (27° 29′ 12″ S, 48° 21′ 30″ W). (Fig. 1), considered the southern limit of tropical fauna, exposed to seasonal cold waters (Segal et al., 2017). We monitored decapods inside and outside a marine no-take protected area (MPA). The three sites inside the MPA included part of Arvo. (Saco D'agua and Rancho Norte sites) and Dese., while three other sites, two in Arvo. (Baia do Farol and Saco do Capim) and one in Aran., were outside the MPA. Highlighting the Arvoredo archipelago houses the only MPA in the entire subtropical rocky shore ecosystems (Anderson et al., 2019). This studied area is near Florianopolis (Santa Catarina Island), the largest urban center in the region (~1.2 million inhabitants). Aranhas island and the open area on Arvoredo island are free to access by these inhabitants. Meanwhile, Deserta Island and the protected areas on Arvoredo are within the monitored and protected MPA, with controlled access. However, this MPA is located in one of the largest fishing grounds in Brazilian waters, an area which encounters frequent issues with illegal fisheries (Anderson et al., 2019), and which has a long history of top predators which have disappeared from the region's records (Segal et al., 2017). Despite these pressures, the MPA is still considerably effective in protecting marine life, and the target fishing grouper *Epinephelus marginatus* (Lowe, 1834), for example, presents higher biomass and larger specimens within the MPA than outside it (Segal et al., 2017). Regarding benthic coverage, the results of the Maare project (Segal et al., 2017) have demonstrated that the areas outside the MPA present higher dominance of filamentous algae turfs, as well as decreased diversity in natural coverage (algae and zoanthids).

2.2. Monitoring methods

The surveys in both study areas were performed by the same decapod taxonomist and the methodology was based in his previous study (Giraldes et al., 2017; 2015a, 2012). The surveys in the coastal coral reef (Po.Ga. and Tam.) were conducted at night, as most decapods are nocturnal. The surveys in the southern rocky shore reef were conducted during the day, the active time for the Mobile Invertebrate Feeder fishes, the main predator of decapods. Therefore, in the coral reefs, we evaluated the exact composition of decapods, observing them during their active time when most fish predators are not active. Conversely, in the rocky reef, we evaluated the decapods that could be observed even during the active time of their predators.

At all sites, a 20-m belt transect was used, and the decapods present within the belt transect area were visually identified. In this study, we specifically searched for the species present in the external habitat (exposed to predators) and crevicular cryptic habitats like cavities, fissures, and caves (in the dark hidden from predators) (Fig. 2), as these are the two main habitats for decapods in reef environments (Giraldes et al., 2017). The observed species were noted on diving slates and categorized according to their position in the substrate (external or crevicular).

Twelve replicates were performed at each site in the tropical reef area (i.e., 12 at Tam. and 12 at Po.Ga.), with the diving surveys performed in the same month in the same lunar moment (full moon). For each studied reef bench (Po.Ga. and Tam.), we selected 4 very shallow reef areas (0–2 m), 4 shallow reef areas (2–4 m), and 4 deeper reef areas (greater than4m). The monitoring process for the tropical reef followed the description of (Giraldes et al., 2017), observing all decapods in the reef edge from a vertical perspective, from the surface to the bottom, and recording the species position according to the located habitats. Here, one vertical sampling area was created per transect, with a length of 20 m of reef edge, and a width based on the reef edge depth (sampling area of 20 m × site-depth).

In the case of the rocky reef, it was employed the traditional belt transect method with 20 m of rock shore and 1 m for each side of the transect (sampling area of 20 m × 2 m). Due to expected bias on account of the different decapod compositions in different depths, tidal zones, and hydrodynamics (Giraldes et al., 2015a, 2012), we performed 1 transect in shallow areas (1–2 m), and a second transect in the same rock shore area but at a greater depth, with the transect line just above the interface between the rock and the soft substrate. Most sites in the present study had an interface of between 5 and 8 m. This monitoring process was based on the protocol designed for the Maare project for the purpose of monitoring decapods in this ecotype of rock shore environment (Giraldes and Freire, 2014). In this study, we selected 4 rock reef areas (20 m) in each one of the sample sites (3 sites/island within the MPA, and 3 outside), totaling 12 rock shore areas (20 m long) inside the MPA, and 12 outside. Considering 2 transects per rock shore area, a total of 24 transects were performed inside the MPA, and 24 outside.

2.3. Species categorization

The decapods were categorized based on size, dominance and use for human consumption. Large decapods (larger than ~10 cm), traded in local markets, such as lobsters and crabs, they are considered as target species for fishing and human consume. The dominant species with small sizes (smaller than ~5 cm) they are considered the main biomass of small prey for fish (Mobile Invertebrate Feeder). The general biodiversity of small size decapods, with less representativeness in the ecosystem, they were evaluated separately as less-representative species.

Hermit crabs were not considered small prey for fishes because despite the small sizes of some species their evolutionary defense strategy, protect them inside the gastropod-shell and only when swapping shells or when within smaller shells became vulnerable for predators (Bertness, 1982). The fish cleaner species like the banded shrimps, also were not included as small food for fishes because their natural interspecific association with fishes (Saito et al., 2009).

2.4. Data analysis

The ecological analysis was based on the abundance (number of specimens), richness (number of species) per transect and dominance (percentage of each species according with the total species per site). The average number of specimens of each species (or group of species) per transect, per area, and per habitat (external and crevicular) are presented. The statistical divergences between the abundances and

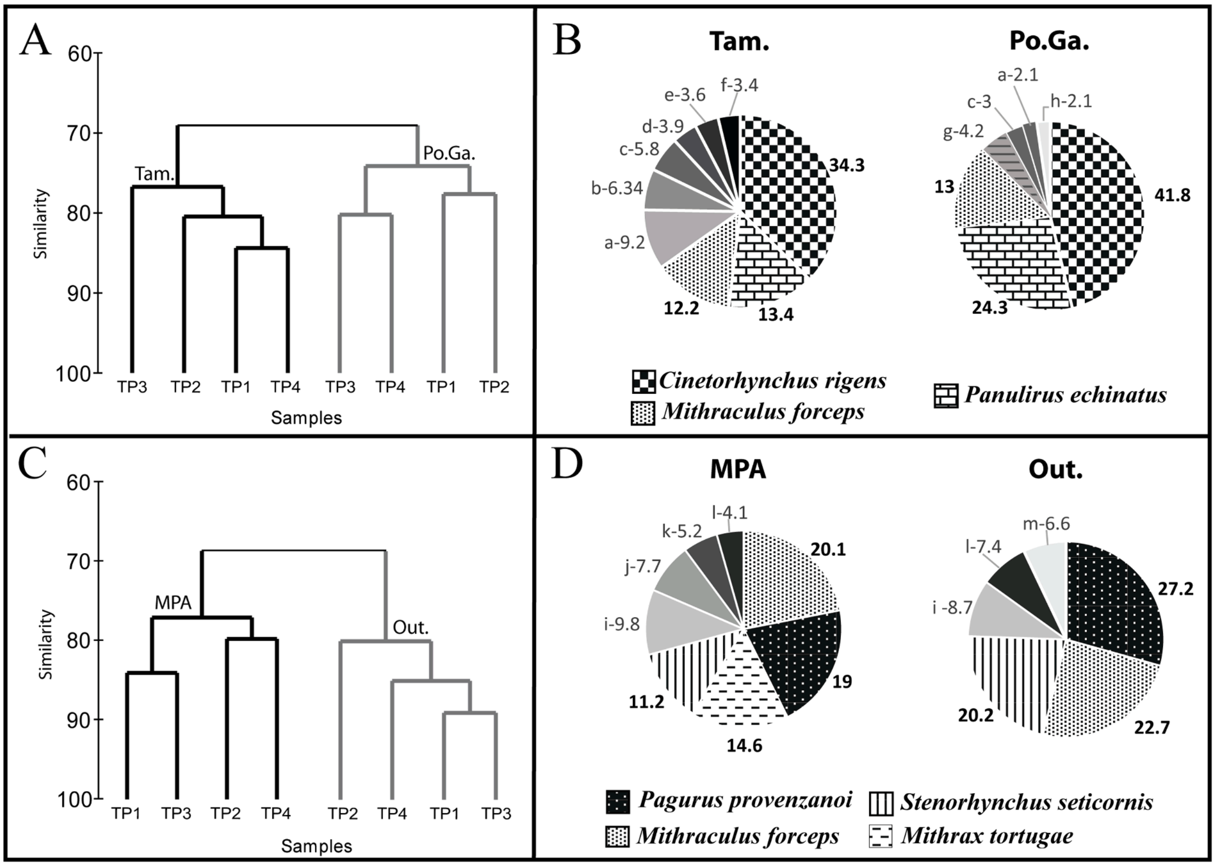

Fig. 3. Results of multivariate analysis at (A, B) Costal Coral Reefs, comparing Porto de Galinhas (Po.Ga.) and Tamandare (Tam.); and at (C, D) Rocky Reefs, comparing sites within the MPA with sites outside (Out.). Clusters of ANOSIM [A and C]; and charts for SIMPER [B, D] show the dominant species (bold) and the exact contribution (%) per species, (a) *M. brasiliensis*, (b) *C. tibicen*, (c) *S. hispidus*, (d) *N. acuticornis*, (e) *P. gundlachi,* (f) *B. biunguiculatus*, (g) *J. antiguensis* (h) *P. provenzanoi*, (i) *M. nodifrons*, (j) *L. ankeri*, (k) *D. insignis*, (l) *D. hispidus* and (m) *C. ruber*.

richness were evaluated using the Mann-Whitney non-parametric test (Zar, 1999) with a significance level of 5%.

Using Primer6 Software (under BWG permission), a multivariate analysis was performed, comparing the sites within the protected areas with the sites outside the protected areas. Four pools of three reef areas were generated, and the averages of the results used for comparison. The pools were created on account of the bias relating to decapod composition in sites of different depths, relating to hydrodynamic and depth relation (Giraldes et al., 2012). In the tropical coral reefs in each pool, we selected 1 transect in very shallow, 1 in shallow, and 1 in deeper reef edges (3 coral reef edges per pool). Similarly, in the subtropical rock reefs, we created a pool using 1 rock reef area (20 m) per studied site/ island, with 3 rock reef areas per pool, in order to avoid the bias relating to hydrodynamic and site specificity. For the ANOSIM all data were transformed (square root), standardized (samples by total), reassembled (Bray-Curtis similarity) and clustered, producing a cluster chart for highlighting the similarities. Using the same dataset, a SIMPER analysis was performed highlighting the similarity percentages and the specific species contribution in the population in each area.

3. Results

3.1. Multivariate analysis

The clusters in the ANOSIM demonstrate the existence of different decapod composition/proportion according with anthropogenic pressures, as in the coral reefs (Fig. 3A) as in the rock reefs (Fig. 3C). The SIMPER analysis illustrates the contribution (%) of each species per site following the ANOSIM pattern. Considering the decapods from coral reefs (Fig. 3B) in Po.Ga. the 3 dominant species controlled almost 80% of the entire population and only 4 extra species contributed with the remains 20% (total of 7 species); and in the less threatened site (Tam.) around 60% was dominated by the 3 dominant species and extra 6 species contributed with the remained 40% (total of 9 species). The same contribution pattern in the SIMPER was observed in the rock reefs (Fig. 3B), with 3 dominant species controlling around 75% in sites outside the MPA and only 3 specie with the remaining 25% (total 6 species), while within the MPA sites 4 species dominated around 60% and 4 species the remaining 40% (total 8 species).

3.2. Abundances per sites and habitats

With an average of 108 specimens/transect, Po.Ga. has a significantly higher number of specimens ($p = 0.03$), compared with Tam. with 71 sps/tran. (Fig. 4A). In the External-Habitat, there was an average of 40.2 sps/tran. in Po.Ga.and 26.2 sps/tran. in Tam. ($p = 0.091$) (Fig. 4B). In the Crevicular-Habitat, there was an average of 64.7 sps/tran. in Po.Ga. and 45.3 sps/tran. in Tam. ($p = 0.078$) (Fig. 3B). These results suggest there were significantly more decapods in the highly threatened sites.

On the Rock Reefs (Fig. 4 C-F) the same pattern was recorded. At Arvoredo Island, outside the MPA an average of 22 specimens/tran was recorded and inside the MPA an average of 10 spp/tran ($p = 0.001$) (Fig. 4C). In the External-Habitat. (Fig. 4D), 15 spp/tran was recorded outside the MPA and 7 spp/tran inside ($p = 0.007$); and in the Crevicular-Habitat 29 spp/tran outside and 13 spp/tran inside ($p = 0.027$). In Aranha/Deserta Island, an average of 24 spp/tran, was recorded outside the MPA and, an average of only 11 spp/tran within

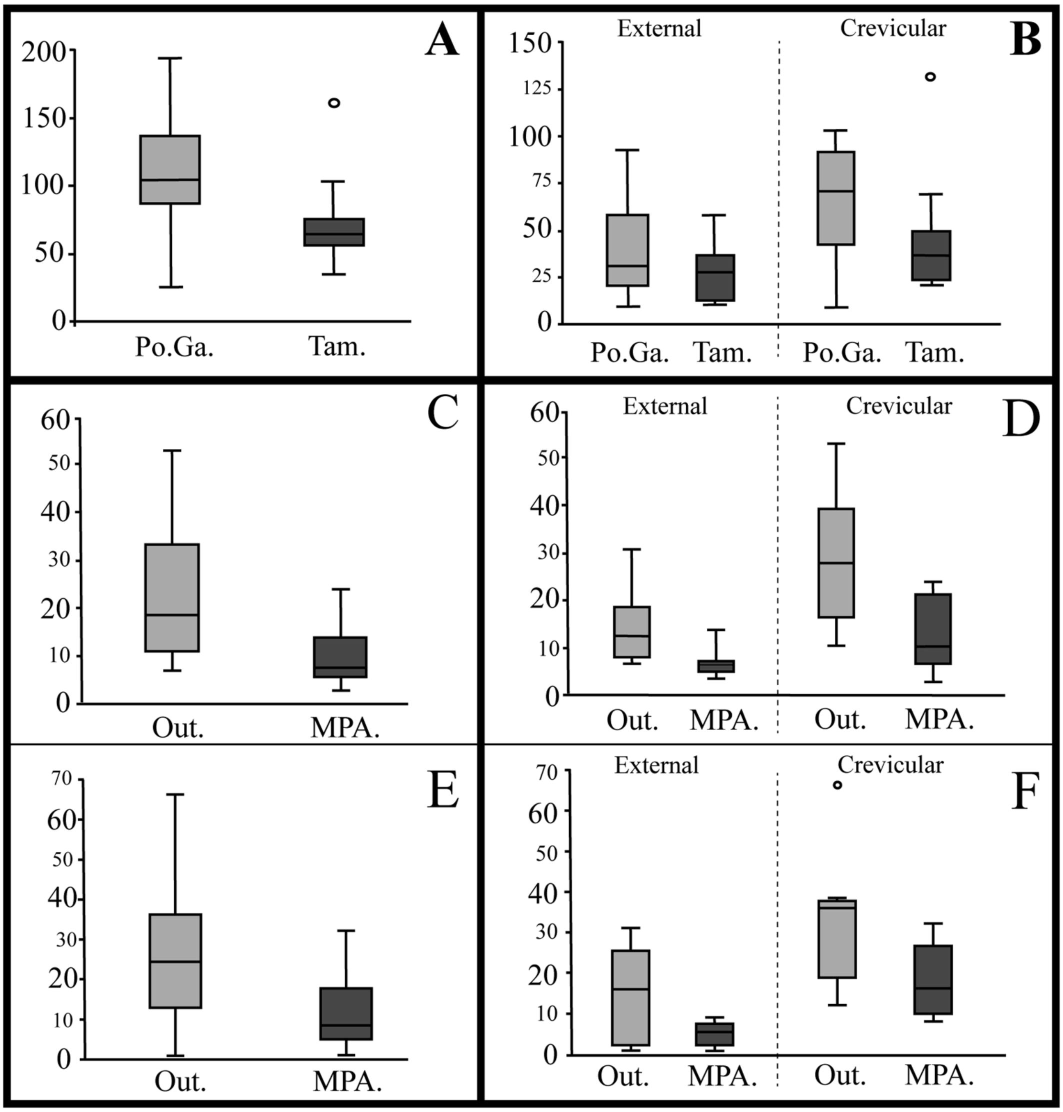

Fig. 4. Abundance of decapods (A, B) at the sites on the tropical coral reef (Po.Ga. and Tam.), and at the sites on the rocky shores (within the MPA and outside it [Out.]), on (C, D) Arvoredo Island and (E, F) Deserta and Aranhas Island. These data consider the entire population per transect [A, C, E], and the populations of the external and crevicular habitats [B, D, F].

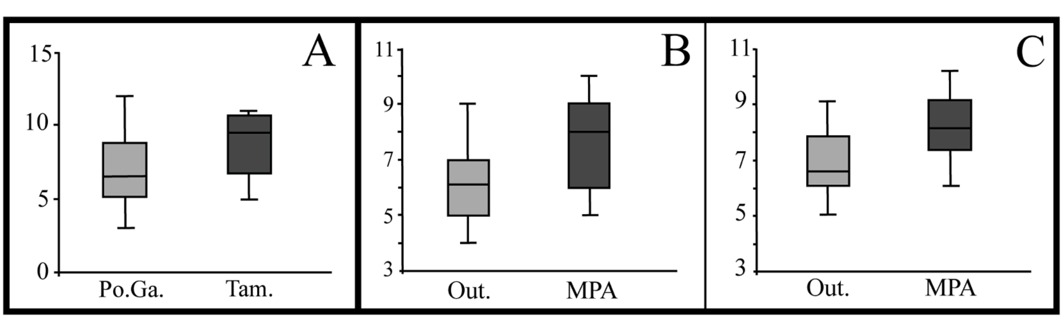

Fig. 5. Richness of decapods (A) at the sites on the tropical coral reef (Po.Ga. and Tam.), and at the sites on the rocky shores (within the MPA and outside it [Out.]), on (B) Arvoredo Island and (C) Deserta and Aranhas Island.

the MPA ($p = 0.025$) (Fig. 4C). In the External-Habitat 15 spp/tran were recorded outside the MPA and only 5 spp/tran ($p = 0.031$) in the MPA sites (Fig. 4D); Crevicular-Habitat follow the pattern (Fig. 4D), sites outside MPA housed 33 spp/tran, and in the MPA an average of 18 spp/tran ($p = 0.040$) was recorded. The abundance was higher in the sites outside the MPA and the external and crevicular habitats follows the same pattern.

3.3. Richness

Differently from the abundance, the richness was higher in less threatened sites, with average of 7 and 9 sps/tran. respectively for Po.

Table 1
Species abundance per transect (average). Decapods recorded on external and crevicular habitats on corals reefs of Tamandaré (Tam.) and Porto de Galinhas (Po.Ga.), Northeastern Brazil. In bold the dominant species. * absent; © Crevicular, Æ external and (š) soft-bottom species according with (Giraldes et al., 2017); $ large species traded in fishing markets; ‡ Small prey for fishes.

Species	External		Crevicular	
	Po.Ga	Tam.	Po. Ga.	Tam.
STENOPODIDEA				
© *Stenopus hispidus* (Olivier, 1811)	1	0,08	0,25	0,92
CARIDEA				
© ‡ ***Cinetorhynchus rigens*** (Gordon, 1936)	**21,42**	**9,58**	**44,83**	**33,75**
Æ ‡ *Brachycarpus biunguiculatus* (Lucas, 1849)	0,25	0,33	*	*
© ‡ *Brachycarpus holthuisi* Fausto Filho, 1966	*	*	0,42	0,17
© ‡ *Janicea antiguensis* (Chace, 1972)	0,08	*	0,91	0,66
ASTACIDEA				
© *Enoplometopus antillensis* (Lütken, 1865)	*	*	0,16	*
ACHELATA				
© *Palinurellus gundlachi* von Martens, 1878	*	0,58	0,08	0,16
© $ ***Panulirus echinatus*** Smith, 1869	**4,5**	**1,33**	**16,75**	**8,16**
© $ *Panulirus laevicauda* (Latreille, 1817)	*	*	*	0,16
© $ *Panulirus meripurpuratus* Giraldes & Smyth, 2016	0,16	*	*	*
Æ $ *Parribacus antarcticus* (Lund, 1793)	0,16	0,08	0,08	*
ANOMURA				
Æ *Calcinus tibicen* (Herbst, 1791)	0,42	0,83	*	0,42
© *Cancellus ornatus* Benedict, 1901	*	*	0,33	0,16
(š) *Dardanus venosus* (H. Milne Edwards, 1848)	*	0,08	*	*
© *Paguristes erythrops* Holthuis, 1959	*	*	0,25	0,25
Æ *Pagurus provenzanoi* Forest & de Saint Laurent, 1968	4.1	0,92	*	*
BRACHYURA				
Æ *Dromia erythropus* (Edwards, 1771)	*	0,16	*	*
Æ $ *Menippe nodifrons* Stimpson, 1859	0,25	0,16	0,33	*
Æ ‡ *Microphrys bicornutus* (Latreille, 1825)	0,16	0,25	*	*
Æ ‡ ***Mithraculus forceps*** A. Milne-Edwards, 1875	**9**	**4,75**	*	*
Æ ‡ *Mithrax braziliensis* Rathbun, 1892	0,5	3,75	0,16	0,08
Æ ‡ *Mithrax hemphilli* Rathbun, 1892	0,16	0,25	*	*
© $ *Damithrax hispidus* (Herbst, 1790)	0,58	0,25	0,08	0,42
Æ ‡ *Nemausa acuticornis* (Stimpson, 1870)	*	0,58	*	*
Æ ‡ *Domecia acanthophora* (Desbonne, in Desbonne & Schramm, 1867)	0,33	0,58	*	*

Ga. and Tam. (p = 0.055) (Fig. 5A). The same recorded in the Rock Reef seascape, with higher richness within the MPA; at Arvoredo Island an average of 8 sp/tran inside and 6 sp/tran outside the MPA (p = 0.0042) (Fig. 5B); and in Deserta/Aranhas Island, an average of 8.1 sp/tran was recorded inside the MPA and in 6.7 sp/tran in the outside sites (p = 0.0472) (Fig. 4C). Overall, these results demonstrate the high degrees of richness in the less threatened sites.

3.4. Species composition in the tropical coral reef

The main dominant species in the tropical species was “small prey for fishes” (Fig. 3B) (Table 1). More specifically the shrimp *C. rigens* (Fig. 6A) and the crab *M. forceps* (Fig. 6B). The shrimp *C. rigens* was significantly more abundant in more threatened sites (p = 0.05), in special in the External-Habitat (p = 0.03) (Fig. 6 A) outside its natural refuge. *Mithraculus forceps,* was also significantly more abundant in the threatened site Po.Ga. (p = 0.05) (Fig. 3F) in the only habitat where it occurred in this reef. The less representative species together (Fig. 6C) they were significantly more abundant in the less impacted sites (Tam.) (p = 0.036).

The exception was *Enoplometopus antillensis*, *Brachycarpus holthuisi* and *Janicea antiguensis* (Table 1) all recorded preying on *C. rigens* (Personal observation) and presented higher abundance in Po.Ga., where their prey was more abundant. The same was recorded for the target and bycatch species in the fishing market, that were less representative in Tam., the site with active lobster fishing effort, such as the lobsters *Panulirus echinatus, Panulirus meripurpuratus, Parribacus antarcticus* and the large crabs *Menippe nodifrons* and *Damithrax hispidus.* Comparing the composition (Table 1) the sites with less threat history (Tam.) presented more species in total, headed by brachyuran crabs within the External-Habitat.

3.5. Species composition in the subtropical rocky reef

Dominant species in the Subtropical Rock Reef presented higher abundance outside the MPA, such as *Mithraculus forceps* (Fig. 7A) with average of 9.3 sp/tran outside and 5.6 inside the MPA (p = 0.0461); *Pagurus provenzanoi* (Fig. 7B) with an average of 14.6 and just 7.6 within the MPA (p = 0.00086); and the spider crab *Stenorhynchus seticornis* (Fig. 7C) with an average of 14 sp/tran and 7.2 in the sites within the MPA (p = 0.00213). The same was recorded with the abundant crevicular shrimp *Lysmata ankeri*, with higher abundance in the sites with higher anthropogenic pressure (Table 2).

The less-representative species (Fig. 7D), they were more abundant inside the MPA, with average of 10 sp/tran and average of 4.8 outside (p = 0.0026). Some less-representative species were recorded only within the MPA such as *Platypodiella spectabilis* (Fig. 7D *II*), *Hypoconcha parasítica* (Fig. 7D *III*) and *Petrochirus diogenes* (Fig. 7D *IV*).

Concerning the economically target species, *Panulirus laevicauda, Menippe nodifrons* and *Damithrax hispidus* they were more representative, or exclusively recorded, in the sites within the MPA (Table 2). The opposite occurred with the scavenger *Cronius ruber,* which partake among the dominant in sites outside the MPA (Fig. 3D). The species composition was higher in the areas within the MPA, with a total of 16 species, and outside the MPA only 12 species was recorded (Table 2).

4. Discussion

4.1. Reef-decapods as bioindicator of threatened ecosystems

The results in this study clearly demonstrated the existence of a population plasticity of reef-decapods with the anthropogenic pressure that a given reef settings was historically exposed. Several ecological patterns were recorded within this plasticity, such as: higher degrees of richness in the “less-threatened” sites and higher abundance in the threatened sites; a higher abundance of the naturally dominant small prey species; this small prey species also expose themselves in external habitats with more intensity in the threatened ecosystems; reef-decapods with economic importance are less representative in anthropogenically impacted sites; close associations between reef-decapods and pristine ecosystems, particularly in species associated with corals and further colonial benthic cover.

Regarding the higher abundance of small prey decapods in threatened reef settings, it is possible to assume that these prey resource they do increase in sites with heavier anthropogenic pressure. In reef sites that are recorded as natural habitat for several predators like the fishes grouped in the Mobile Invertebrate Feeder (MIF) category (Freitas et al., 2017; Randall and Bishop, 1967; Sonnefeld et al., 2014; Wainwright and Richard, 1995). Highlighting that the mouth size of MIF fishes is related

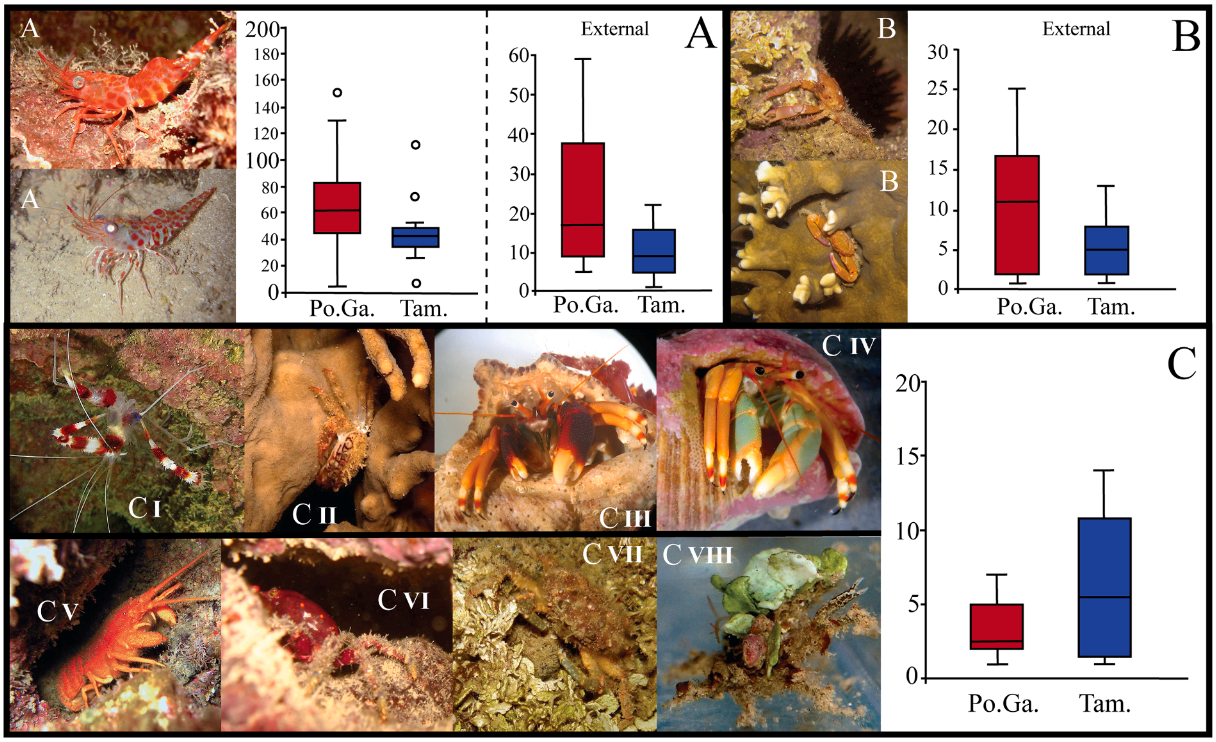

Fig. 6. Abundance (per transect) of decapods on the tropical coral reef (Po.Ga. and Tam.). (A) *Cinetorhynchus rigens* total data only on external habitat; (B) *Mithraculus forceps* on external habitat, (C) all less-representative species together, including S. hispidus [CI], *D. acanthophora* [CII], *C. tibicen* with brown morphotype [CIII], and green morphotype [CIV], *P. gundlachi* [CV], *M. brasiliensis* [CVI], *M. hemphilli* [CVII], and *M. bicornutus* [CVIII]. Credits: Claudio Sampaio (Buia) [CV]; Bruno W. Giraldes [all others].

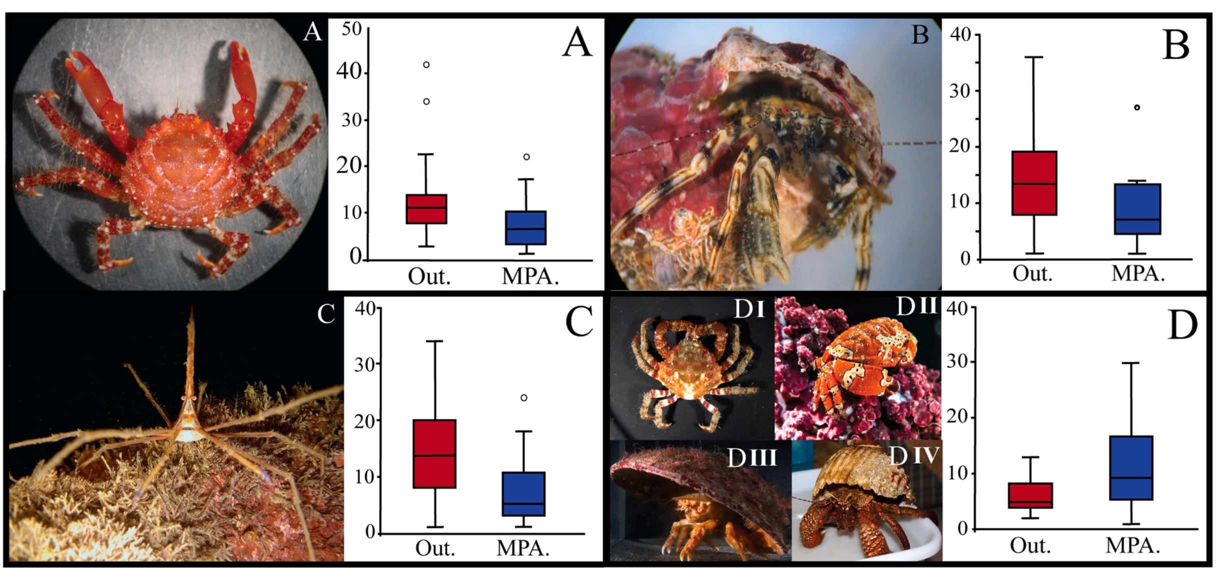

Fig. 7. Abundance of representative decapods on rocky reefs, comparing sites outside (Out.) and inside the Marine Protected Area [MPA]. (A) *M. forceps*, (B) *P. provenzanoi*, (C) *S. seticornis*; (D) all less-representative species together, highlighting *M. tortugae* [DI], *P. spectabilis* [DII], *H. parasitica* [DIII], and *P. diogenes* [DIV].

to their prey size (Artero et al., 2015; Gibran, 2007; Sonnefeld et al., 2014; Wainwright and Richard, 1995) and consequently the small decapods can feed even small or juvenile fishes. Considering that in the historical sequence of overfishing the large fish species are removed first (like large groupers), and that the MIF fishes (like groupers) are reduced in more threatened and easily accessible shallow sites (Anderson et al., 2019; Engmann, 2006; Ferreira et al., 2004; Pereira et al., 2018; Segal et al., 2017) it is possible to consider that the high abundance of small prey decapods might be related to the top-down trophic imbalance with the removal of predators in overused reef sites.

An important concern in this prey-predator relation is regarding the visual location of preys performed by MIF fish (Artero et al., 2015; Gibran, 2007), because avoiding visual identification is an evolutionary strategy used by most decapods. Similar to those of insects in forests for avoiding visual identification by birds (Umbers et al., 2014). Indeed, most decapods have evolved specific camouflage strategies to avoid visual identification (Duarte et al., 2017; Giraldes and Smyth, 2016; Palma and Steneck, 2001; Stevens et al., 2014). With disruptive

Table 2
Species abundance (average) per transect/rock-shore (20 m of shoreline). Decapods recorded on external and crevicular habitats on subtropical rock reefs, Southeastern Brazil. Comparing inside the Marine Protected Area (MPA) and outside (Out). Dominant species in bold. * absent; © Crevicular, Æ external and (š) soft-bottom species; $ large species traded in fishing markets; ‡ Small prey for fishes.

Species	Arvoredo Island				Deserta and Aranhas Islands			
	External		Crevicular		External		Crevicular	
	MPA	Out	MPA	Out	MPA	Out	MPA	Out
STENOPODIDEA								
© *Stenopus hispidus* (Olivier, 1811)	*	*	*	*	*	*	0.38	*
© *Stenopus spinosus* Risso, 1827	*	*	*	*	*	*	*	0.25
CARIDEA								
© ‡ ***Lysmata ankeri*** Rhyne & Lin, 2006	*	*	**1.3**	**2.3**	*	*	**3**	**7.75**
ACHELATA								
© $ *Panulirus laevicauda* (Latreille, 1817)	0.06	*	0.06	*	*	*	0.50	*
ANOMURA								
Æ ***Pagurus provenzanoi*** Forest & de Saint Laurent, 1968	**3.94**	**11.06**	**1.19**	**1.81**	**4**	**12.13**	**2**	**0.75**
(š) *Dardanus insignis* (de Saussure, 1857)	*	*	0.94	0.13	*	*	1.3	0.95
(š) *Petrochirus diogenes* (Linnaeus, 1758)	*	*	0.06	*	*	*	*	*
Æ *Calcinus tibicen* (Herbst, 1791)	0.31	*	0.5	*	0.63	*	0.5	0.63
Æ *Paguristes tortugae* Schmitt, 1933	0.19	*	0.88	0.5	*	0.38	*	**4.1**
BRACHYURA								
Æ ‡ ***Stenorhynchus seticornis*** (Herbst, 1788)	**0.15**	**0.06**	**2.80**	**8.44**	**0.38**	**2.38**	**9.38**	**15.63**
© $ *Damithrax hispidus* (Herbst, 1790)	*	*	1.06	1.44	*	*	1.13	0.88
Æ ‡ ***Mithraculus forceps*** A. Milne-Edwards, 1875	*	**0.06**	**6.69**	**12.5**	**0.13**	**0.13**	**7.5**	**10**
© ‡ *Mithrax tortugae* Rathbun, 1920	*	*	3.56	0.31	*	*	1.38	1.38
Æ $*Menippe nodifrons* Stimpson, 1859	0.06	*	1.88	1.44	*	*	0.5	*
Æ *Platypodiella spectabilis* (Herbst, 1794)	*	*	*	*	0.75	*	*	*
(š) *Cronius ruber* (Lamarck, 1818)	*	*	0.31	1.3	*	*	0.19	3.25
(š) *Hypoconcha parasitica* (Linnaeus, 1763)	*	*	*	*	*	*	0.06	*
Æ ‡ ***PORCELANIDAE***	*	0.06	0.2	0.13	*	*	0.25	0.13

camouflage among the benthic coverage in external habitats and with a crevicular camouflage (reddish color) to avoid visual identification in dark environments (Giraldes et al., 2017). Therefore, the visual identification of a decapod by divers in this study means that they have failed in their evolutionary camouflage strategy or that there are not enough predators to control the number of those small decapods. This strongly corroborates with the statement that the great presence of small decapods (0.5–10 cm) in threatened sites is due to the long-term (historical) absence of predators. Driving to the conclusion that the visual identification of abundant small decapods, can be used to illustrate the top-down effect of unrecorded historical overfishing.

Another ecological pattern regarding the dominant species is that they found in threatened sites abundant and available feeding resource. Highlighting that most of those dominant species recorded here, they are traded in aquarium industry as ""cleaning crew" from unbalanced environment. For instance, *Cinetorhynchus* species has the ability for controlling the biofouling coverage in hard substrates a considered pest in threatened marine areas (Dumont et al., 2009). *Mithraculus* has a famous potential for cleaning and controlling the excessive microalgae in eutrophic environments (Calado, 2006; Penha-Lopes et al., 2007). *Lysmata* is also a famous functional species for controlling pests (Calado and Narciso, 2005). The spider crab *Stenorhynchus seticornis* has an omnivorous/scavengers diet, eating dead marine organism in organically polluted environments (Antunes et al., 2018). This study also recorded higher dominance and abundance of efficient scavenger decapods as small Paguridae hermit crabs (*P.provenzanoi* and *P.tortugae*) (Ramsay et al., 1997) on threatened sites at rocky reefs; and a higher presence of the swimming crab *Cronius ruber* in the highly threatened sites (Duarte et al., 2010; Paul, 1981; Watling et al., 2013). One of the main characteristics of anthropogenically impacted sites is the excessive input of organic matter from the waste of fishing and tourism, wastewater, and the mortality of benthic coverage on hard substrates (Duarte

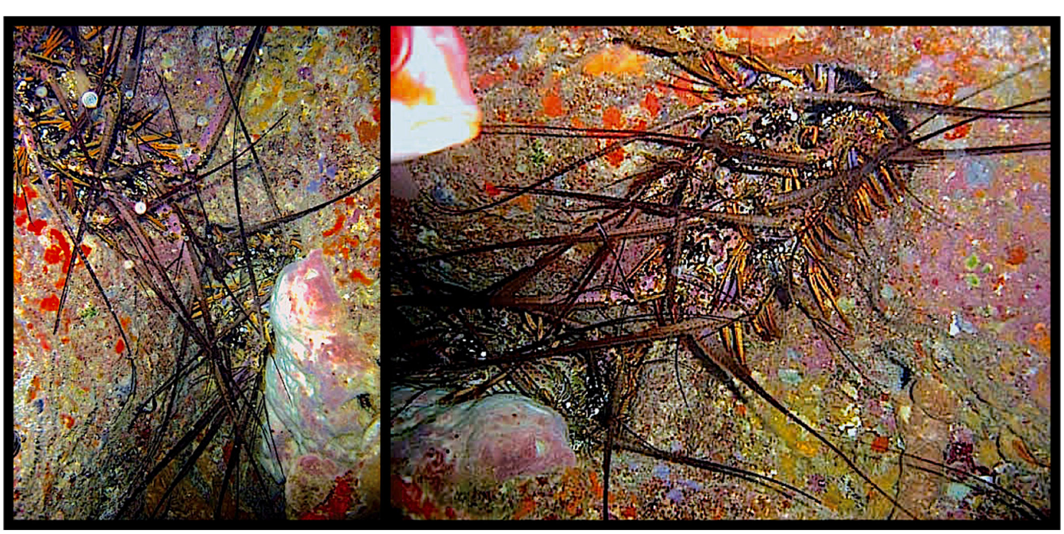

Fig. 8. Spiny lobster Panulirus meripurpuratus (Giraldes and Smyth, 2016), photographed during the 90 s in the tropical reefs in northeast Brazil. Credits: Carla Malucelli.

et al., 2010; Feitosa et al., 2012; Grall and Chauvaud, 2002; Moss, 2011). Certainly, this eutrophication (Moss, 2011) and excessive input of organic matter will directly affect the benthic community, increasing the algae proliferation and nuisance/pest organisms, removing fragile benthic coverage (corals), and consequently increasing the number of scavengers (Grall and Chauvaud, 2002) and further natural "cleaning crew". Considering the efficiency of this decapods for controlling unbalanced microcosmos at marine aquariums, eating algae, nuisance/pest organisms and scavenge on detritus (Antunes et al., 2018; Butler and Mojica, 2012; Calado, 2006; Calado and Narciso, 2005; Penha-Lopes et al., 2007). It is possible to speculate that the significant presence of the cleaning crew decapods in natural ecosystems could be related to the excessive organic matter there (Barradas et al., 2012; Santos et al., 2015), or the proliferation of microalgae or pests/parasites. In other words, the excessive number of those functional reef-decapods might be an indicative for ecosystems in collapsing.

This study also demonstrated that some decapods are absent or with a decreased abundance in threatened sites. For instance the niche specific decapods with close association with benthic coverage (e.g.corals), that are more dependent of the trophic balance stability. Several species, which had been previously recorded as closely associated with benthic coverage (Giraldes et al., 2017, 2015a; Melo, 1996), were absent in this study at the sites considered to be under threat (Table 1 and 2), or present in reduced numbers. Highlighting that those decapods are several times responsible for the survival and protection of those hosts (Boudreau and Worm, 2012; Glynn and Enochs, 2011; Stewart et al., 2006). Similar to the reduction of insect diversity with the removal of forests recorded worldwide (Sánchez-Bayo and Wyckhuys, 2019).

Also in negative relation, the target reef-decapods traded in fishing market are considered the main bioindicator from threatened sites, in special the lobsters due their high associated value per unity. In this study and in both ecoregions (Table 1 and 2), all lobsters and large crabs (including hermit crabs) were either absent in the overused reef sites, or their populations were seen to have decreased. The same target and bycatch fauna reported as affected by the gillnet fishing in previous study (Giraldes et al., 2015b). A tragic populational status is facing one of the main fishing resources in Brazil *Panulirus meripurpuratus,* which was virtually not recorded in several scuba diving in two different ecoregions and in 6 reef areas. An endemic species virtually restricted to Brazil, which uses the studied shallow coastal reefs as a nursery zone during its juvenile stages (Giraldes and Smyth, 2016). Conversely, only 30 years ago, in a shallow marine area in the same reef ecosystem, more than 10 specimens in a single cavity (approximately 1 m^2 of reef area) were recorded during the daytime (Fig. 8). In other words, we strongly suggest the use of *Panulirus meripurpuratus* as main bioindicator of overfishing sites in the threatened Brazilian coastal reefs.

A similar study, which compared decapod assemblages in threatened and pristine coral reefs in the Caribbean Sea, recorded comparable results by including the same species (congeners) in the same concentration. A congener of *Mithraculus* (in a niche equivalence) was the dominant decapod in threatened ecosystem, while in the pristine coral reef site *Domecia acanthophora, Calcinus tibicen* and *Palinurellus gundlachi* were also more representative (González-Gómez et al., 2018). Demonstrating that the results recorded in this study is an ecological pattern, with target groups with niche equivalence and human use, reflecting positively or negatively over the anthropogenic pressure.

5. Conclusion

5.1. The Ghost of the past anthropogenic impact

This study demonstrate a pattern reported for most ecosystems in the Anthropocene, with the existence of an impact gradient that follows the frequency of human access and the proximity to large urban centers; the easier the access for humans, the higher the impact (Al Maslamani et al., 2018; Jackson et al., 2001; Mirabella and Allacker, 2017; Pereira et al., 2018). Indeed, here we recorded the population plasticity of some target reef-decapods basically reflecting the past bottom-up and top-down impacts. Demonstrating that Rapid Assessment Survey (RAS) using Underwater Visual Census (UVC) methods for monitoring reef-decapods can be used for illustrating the past anthropogenic impact and for classifying the collapse level of a given site in threatened ecosystem.

In a theoretical point of view, and considering that: 1) in the Anthropocene, it was inserted the human population as a very predatory competitor in the natural ecological relations within the marine ecosystems; 2) the resilient biodiversity that inhabits threatened or collapsed ecosystem, will naturally occupy the vacant niche, and will find the populational balance according with the available resource and the populational balance according with the prey-predator proportions. In this way, in an analogous concept to the theory known as "Ghost of Competition Past" (Connell, 1980) we are proposing the use of reef-decapods as target ecological indicator for highlighting the "Ghost of the Past Anthropogenic Impact" and determine anthropogenic impact level in reef areas. A concept that can be used for the future classification of collapsed and threatened ecosystem in environmental management plans.

CRediT authorship contribution statement

Bruno Welter Giraldes: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing. **Petrônio Alves Coelho:** Conceptualization. **Petrônio Alves Coelho Filho:** Formal analysis, Supervision. **Thais P. Macedo:** Data curation, Formal analysis, Validation. **Andrea Santarosa Freire:** Conceptualization, Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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